

Sign and Orientation Issue in LB and YFB

OpenAI Codex

2026-05-08

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1 Purpose

This note explains the sign/orientation issue in the LB and YFB models.

The main goal is to separate three different ideas that can easily get conflated:

1. a **true mathematical sign indeterminacy** in the factorization
2. a **risk-score orientation convention** for Cox prediction
3. the **current implementation fix**, which flips the sign of the fitted survival coefficients when the training concordance is below 0.5

The key point is that the current sign flip is not changing the fitted subspace or inventing new signal. It is choosing an orientation for a predictor whose sign is not fully identified by the factorization alone.

2 The Core Mathematical Issue

Both LB and YFB inherit sign indeterminacy from matrix factorization.

2.1 LB

The genomics model is

$$Y \approx LF^\top.$$

For any factor k , if we flip the sign of both:

$$l_{\cdot k} \rightarrow -l_{\cdot k}, \quad f_{\cdot k} \rightarrow -f_{\cdot k},$$

then the product is unchanged:

$$(-l_{\cdot k})(-f_{\cdot k})^\top = l_{\cdot k}f_{\cdot k}^\top.$$

So the reconstruction does not determine the sign of a factor uniquely.

2.2 YFB

YFB still uses the same reconstruction

$$Y \approx LF^\top,$$

so the same factor-sign indeterminacy remains.

In addition, the YFB predictor uses

$$Z_F = YF.$$

If a factor column flips sign,

$$f_{\cdot k} \rightarrow -f_{\cdot k},$$

then the corresponding projected score also flips:

$$Z_{F,\cdot k} \rightarrow -Z_{F,\cdot k}.$$

So YFB inherits the same orientation ambiguity, just expressed through the projected score rather than through latent L .

3 Why Survival Does Not Automatically Fix the Sign

At first glance, it may seem like the Cox model should determine the correct sign, because hazard depends on the linear predictor.

But in both LB and YFB, the relevant factor/sign pair can be re-expressed with an equal and opposite coefficient sign:

3.1 LB

If

$$\eta = L\beta,$$

then flipping one factor column and its corresponding coefficient gives

$$(-l_{\cdot k})(-\beta_k) = l_{\cdot k}\beta_k.$$

So the predictor is unchanged.

3.2 YFB

If

$$\eta = Z_F\tilde{\beta},$$

and

$$Z_{F,\cdot k} \rightarrow -Z_{F,\cdot k},$$

then

$$(-Z_{F,\cdot k})(-\tilde{\beta}_k) = Z_{F,\cdot k}\tilde{\beta}_k.$$

Again, the predictor is unchanged.

So the survival model does not eliminate the sign ambiguity by itself. It only identifies the **product**.

4 The Additional Practical Issue: Risk-Score Convention

Separate from factor-sign indeterminacy, there is a simpler convention issue:

- should a **larger** risk score mean **worse** prognosis?
- or should a larger score mean **better** prognosis?

A Cox linear predictor is only meaningful up to monotone interpretation unless the codebase enforces a convention.

In practice, the project wants:

$$\text{larger risk score} \Rightarrow \text{larger hazard} \Rightarrow \text{worse prognosis.}$$

That is a reasonable and standard convention. But it must be applied consistently.

If some code paths evaluate concordance using `risk_score` and others use `-risk_score`, then part of the apparent “sign problem” is really a bookkeeping problem rather than a modeling problem.

5 Why the Issue Appears in Current Fits

The most important implementation-level reason is the initialization and factorization geometry.

5.1 LB

The singular value decomposition initialization and nonnegative truncation can recover a useful low-rank structure while discarding the original biological sign orientation.

So the learned factor may span the right axis of variation but point in the opposite direction.

5.2 YFB

The same issue appears, but now it propagates through

$$Z_F = YF.$$

So if F_k has the opposite orientation from the one that aligns with higher hazard, then the projected score $Z_{\{F, \cdot k\}}$ also has the opposite orientation, and the fitted beta may inherit that reversal.

This is why the sign issue is not “specific to YFB,” though it can be more obvious there.

6 What the Current Code Does

Both fitting functions apply a post-convergence sign correction based on training concordance.

6.1 LB

In `fit_modular.R`, the code computes the training concordance using the final predictor. If concordance is below 0.5, it flips the sign of `EBeta`.

That changes

$$\eta \rightarrow -\eta$$

and therefore changes concordance from approximately

$$C \rightarrow 1 - C.$$

6.2 YFB

In `fit_cox_on_yf.R`, the code does the same thing using the YFB predictor

$$\eta = Z_F \tilde{\beta}.$$

If training concordance is below 0.5, it flips the sign of `EBeta`.

7 Why This Is Not Purely Ad Hoc

The sign flip can feel ad hoc, but there is a principled reason it works:

- the fitted reconstruction/factor space is already in place
- the issue is the orientation of the predictor
- global sign reversal preserves the factor space but changes the hazard ordering

So the current fix is not creating a new model. It is selecting one of two equivalent orientations of the fitted predictor.

That said, the current implementation is still not the cleanest possible formulation.

8 Why It Still Feels Unsatisfying

The discomfort is valid for two reasons.

8.1 1. The orientation rule is external to the model definition

Right now, the model is fit first, then the sign is flipped afterward if concordance is below 0.5.

That makes the orientation decision look like a patch rather than an explicit part of the model.

8.2 2. The convention is not unified everywhere

If cross-validation, benchmarking, and prediction utilities do not all use the same orientation convention, then the sign issue gets mixed up with evaluation inconsistency.

That is one reason the current CV behavior is hard to interpret.

9 A More Principled Formulation

The cleanest future approach would be:

1. define one global convention:
 - larger score means larger hazard
2. treat orientation as part of the fitted model object
3. determine orientation during fitting using one explicit rule
4. apply the same rule in full-data fitting, external prediction, and cross-validation folds

Two reasonable orientation rules are:

9.1 Option A: training concordance rule

Choose the sign so that training concordance is at least 0.5.

This is close to what the current code already does.

9.2 Option B: training Cox objective rule

Choose between `eta` and `-eta` by whichever yields the larger Cox partial log-likelihood on the training set.

This is arguably slightly more model-aligned because it uses the survival objective itself rather than concordance alone.

10 How to Explain This to Advisors

The cleanest explanation is:

- the factorization is sign-indeterminate
- survival depends on the product of score and coefficient, so sign is only identified up to orientation
- the current implementation chooses the orientation that makes the risk score point in the clinically meaningful direction

That is different from saying “we changed a bad result into a good one by cheating.”

What really happened is:

- the low-rank structure was learned
- the prognostic axis was learned
- the sign convention for that axis was not fully fixed by the factorization
- a global orientation choice was then applied

11 Relation to the YFB Derivation

This issue is orthogonal to the main YFB derivation.

The YFB derivation changes:

- what variable enters survival
- how $q(\text{beta_k})$, $q(\text{L_k})$, and $q(\text{F_k})$ are derived

The sign/orientation issue is a separate identifiability and interpretation issue that sits on top of either LB or YFB.

So when explaining YFB, it is helpful to say:

- the derivation defines the model structure
- the orientation issue concerns how the fitted prognostic axis is signed and interpreted

12 Final Summary

The sign/orientation issue has three layers:

1. **Mathematical factor-sign indeterminacy**
 - inherited from matrix factorization
2. **Prediction-orientation convention**
 - larger score should mean larger hazard
3. **Current implementation fix**
 - if training concordance is below 0.5, flip the sign of the survival coefficient vector

So the current sign flip is best viewed as a pragmatic orientation-selection step, not as a change to the learned low-rank structure itself.

The long-term improvement is to make that orientation rule explicit, unified, and part of the fitted model definition.